

Distribution: For public release

A CFD-Based Parametric Study for Fast Prediction of Blast Loading in Urban Environments

Main author^{1*}, co-author², co-author³, Arial 12pt bold

National Building Research Institute, Technion – Israel Institute of Technology, Technion
City, Haifa 32000, Israel

1 organization, country

Keywords: blast wave channelling, street canyon, Hopkinson-Cranz scaling, fast-running engineering model, safety distance, urban blast loading

Abstract

The ability to predict blast behaviour in an urban environment has gained significant interest in recent years. While blast wave propagation in free field can be predicted with reasonable accuracy using well-established empirical and fast-running models, urban environments produce complex phenomena including channelling, shielding, multiple reflections and non-linear interactions between wave fronts, which limit the validity of such simple models. Three-dimensional Computational Fluid Dynamics (CFD) simulations capture these interactions in full, yet their computational cost renders them impractical for broad parametric studies or rapid real-time assessment. In recent years, advances in GPU-based CFD solvers have dramatically reduced simulation runtimes, making it feasible to run a vast number of high-resolution simulations and enabling broad parametric studies that were previously impractical.

This work presents a large-scale parametric study based on CFD simulations using ViperBlast, a GPU-based solver that enables a vast number of high-resolution three-dimensional simulations. The influence of key parameters is examined, including charge location, building dimensions, building density and the aspect ratio of building height to street width, on the pressure and impulse fields throughout the domain.

Preliminary results establish an engineering prediction framework derived directly from the simulation database. For each pressure or impulse level characterised by the scaled distance in free field, a convergence radius was defined beyond which the loading in the urban environment may be assumed equivalent to free field. For the region where the influence of the urban environment remains dominant, two families of equations were fitted for the scaled distance in the urban environment and the corresponding physical radius, enabling rapid prediction of blast loads as a function of geometry and charge weight, and thus direct derivation of safety distances and expected damage. This framework serves as a fast and generalisable decision-support tool for protective design engineers without complex simulations for each scenario.

Introduction

Explosive events in dense urban areas, whether by deliberate attack [1], industrial accident, or legacy ordnance recovered in built-up ground, are becoming more frequent. These events demand a capability to predict the blast loading they impose. This is required for the selective hardening of buildings and critical infrastructure, for safety distances in accidental events and in controlled demolition, and for real-time decisions such as evacuation and area closure. The charge weight and its location are uncertain in such cases, so an order-of-magnitude understanding of the pressures is of greater practical use than a precise value.

Protection against explosive threats has traditionally rested upon free-field knowledge. The peak overpressure and the positive-phase impulse follow from the charge mass W and the standoff distance R alone, through the Hopkinson-Cranz cube-root law [2, 3]. The problem is thereby reduced to the scaled distance $Z = R/W^{1/3}$, in $\text{m/kg}^{1/3}$, and the relations of Kingery and Bulmash [4] express every free-field parameter as a function of Z alone. This description is one-dimensional.

Within a built-up area, that description fails. The wave undergoes reflection from façades, diffraction about corners, channelling along street canyons, and shadowing behind blocks, and the resulting fronts superimpose non-linearly [5]. The geometry enters the definition of the load rather than correcting it, and points at equal scaled distance may sustain markedly different loads. Denny et al. [6] numerically reproduced the trials of Gajewski and Sielicki [7] behind a building corner. With the corner alone, the peak specific impulse at one gauge fell by 38% below the free-field value. Once a further wall was added to the simulation, the impulse instead exceeded that value by 169%, for the same charge standoff distance. The same reversal occurs at the scale of a whole array. Ratcliff et al. [8] report two layouts of identical areal density that differ only in the arrangement of the blocks, one reducing the central peak overpressure by 13.3% and the other increasing it by 35.5%, both relative to the case with no buildings present.

A further departure arises where the charge lies within the street itself. The loading mechanism is then altered: lateral expansion is inhibited, the reflected fronts travel with the incident wave, and geometric attenuation is suppressed [9]. Hopkinson-Cranz scaling was found to hold in a wide street and to break down as the street was narrowed [10]. Ripley et al. [11] termed the regime in which the scaled-distance relations fail the extended near field. They also advanced one quantitative descriptor of enclosure, the degree of confinement. However, it correlates with the average scaled impulse only with appreciable scatter and was intended for ranking scenarios by severity rather than predicting load. A descriptor common to different urban configurations remains unavailable [8].

Computational Fluid Dynamics (CFD) resolves the governing physics, but each simulation returns the load field for a given specific geometry and charge, and not a relation that can be transferred to different cases. One of the fast-running models suggested by Hedhli et al. replaces the flow solution with an analytical superposition of unit waves [12]. Machine-learning surrogates infer the load from a database of computed cases [13]. Both surrender the explicit representation of the coupling between the geometry and the wave. Solvers accelerated by the graphics processing unit (GPU) have been developed in recent years. They can complete in minutes a simulation that requires hours on a CPU-based code [14]. A parametric study of many configurations is thereby brought within reach. This paper reports such a study, conducted in ViperBlast [15] over an idealised periodic urban layout, together with the engineering prediction framework distilled from it.

Methodology

Idealised urban model

The urban layout was idealised as a periodic array of identical rectangular buildings on a regular grid, as shown in Figure 1. Three lengths describe it completely: the square building footprint B , the street width s , and the building height H . The repeating unit cell is of side $B+s$. The buildings were treated as rigid, fully reflecting and impermeable obstacles. This is the prevailing representation in the urban blast literature, and it is conservative for the load channelled along a street [16, 17].

Two detonation scenarios were modelled. In the street detonation (*det1*), the charge lies mid-street between two parallel rows, feeding a single rectilinear canyon with two opposed propagation directions. In the intersection detonation (*det2*), it lies at a four-way crossing and feeds two orthogonal canyons with four such directions. The charge was specified as an equivalent TNT mass W , detonated at ground level.

The charge occupies a point of two-fold symmetry in both scenarios, so one quadrant was modelled. The two symmetry planes and on the ground were fully reflecting, and the remaining

boundaries transmitting. The measured quantities below are accordingly extracted over a 0 to 90 degree interval along the quarter model. The array was continued to the outer boundary of the domain, so the measured field nowhere crosses an edge of the building array. A decay towards the free-field values therefore reflects the weakening of the urban influence and not the end of the array.

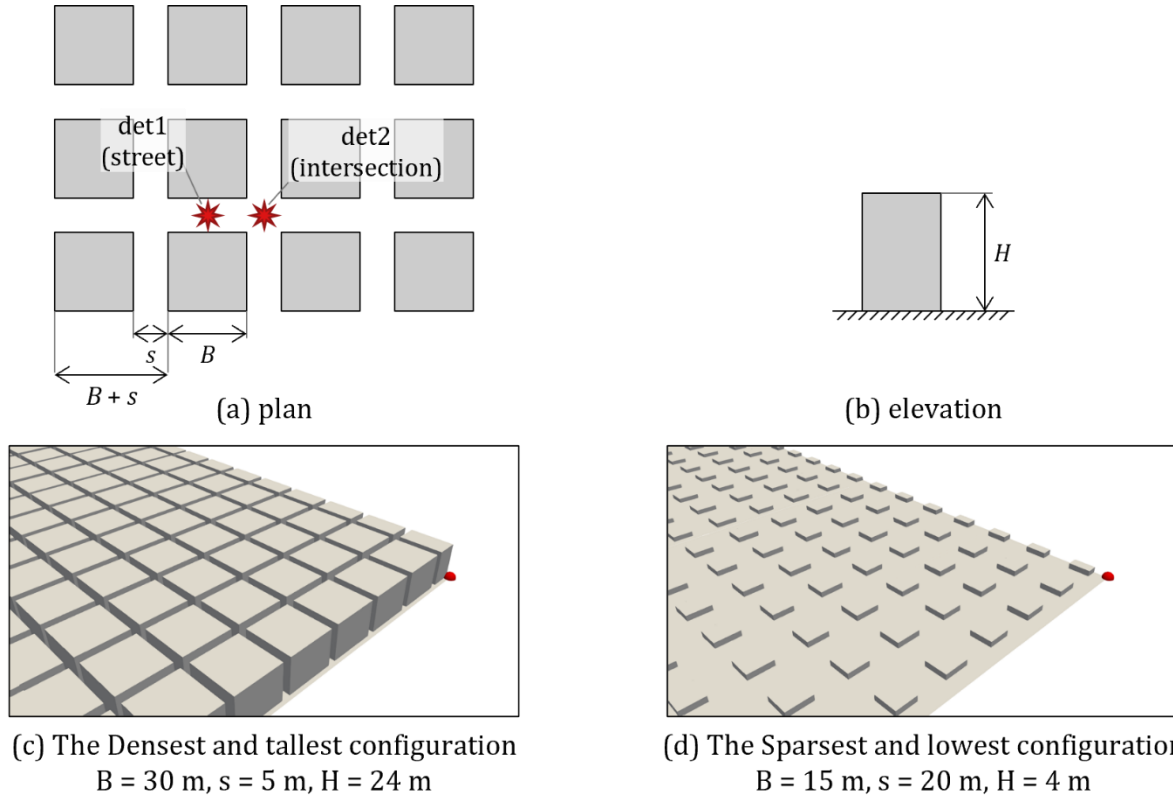


Figure 1. Definition of the idealised urban model. (a) Plan of the periodic array. (b) Elevation. The densest (c) and the sparsest (d) configurations. The charge is marked in red, and one quadrant only is modelled, the two reflecting planes intersecting at it.

Numerical approach

No single grid can resolve both the detonation and the far field at an acceptable computational cost. Each simulation was therefore advanced through a sequence of remapped stages, each solved in the lowest dimensionality its symmetry permits and supplying the initial condition of the next. The sequence comprised up to three stages. The detonation was first solved in one dimension, under spherical symmetry, until the wave reached the nearest façade. For street detonations in the narrow streets ($s = 5$ and 8 m), a two-dimensional axisymmetric stage followed, about an axis through the charge normal to the two façades. It carried the wave along the street until it reached the end of the block or the roof level. In all other configurations this stage was omitted. The solution was finally remapped onto the three-dimensional quadrant model, which was solved on three successive grids of increasing extent, the last reaching 500 m in plan.

The models were solved in ViperBlast, a GPU solver integrating the compressible Euler equations explicitly on a structured Cartesian grid. The scheme is second-order accurate in space and in time. The air was treated as an ideal gas. The free-field predictions of the solver agree with Kingery and Bulmash to within approximately 10% in peak overpressure [18]. The impulse falls consistently some 10% below the empirical curves, an offset attributed to the conservatism of the relations themselves rather than to the solver [18].

The three-dimensional cell sizes of 15, 50 and 100 cm were fixed by a mesh convergence study on the most demanding configuration. Tolerances of 10% on the peak overpressure and 5% on the impulse were met at between 95.4 and 100% of 632 monitoring points. The discretisation was held invariant across all configurations, and each reference run used the same grids as its urban run, so any residual bias is shared by the two members of each pair. This mesh convergence is a property of the discretisation and is not the convergence radius defined below.

Scaled parameter space

The problem was scaled in the manner of the free field, following the Hopkinson-Cranz law. The charge weight enters solely through the Hopkinson length $W^{1/3}$, and every distance is expressed in the scaled form Z . The geometry is then defined by three parameters:

$$\rho = \frac{B^2}{(B+s)^2} ; \quad \tilde{s} = \frac{s}{W^{1/3}} ; \quad \frac{H}{s} \quad (1)$$

Two of the parameters, ρ and H/s , are dimensionless. The scaled street width $\tilde{s}$ carries the units of the scaled distance, $m/kg^{1/3}$. Each parameter corresponds to a physical mechanism. The plan-area density ρ is the built fraction of the ground plane, and it governs the reflecting surface the wave encounters. The scaled street width $\tilde{s}$ expresses the street width relative to the charge, in the same way that Z expresses the standoff distance. At $\tilde{s} = 1 m/kg^{1/3}$, the street width in metres equals the cube root of the charge mass in kilograms. This value is a candidate regime boundary between a street that channels the blast and one that blocks it. The canyon aspect ratio H/s measures the depth of the street channel.

Several street widths were considered (Table 1), where the two extreme street widths were selected to bracket the regime boundary $\tilde{s} = 1 m/kg^{1/3}$: at $s = 5 m$ the larger charges bring $\tilde{s}$ below this value, whereas at $s = 20 m$ it stays above it throughout the weight range. A configuration denotes one simulation, defined by five inputs: B , s , H , W and the detonation scenario. A geometry denotes one combination of B , s and H . The design comprises a balanced core block of 72 configurations, in which every combination of two values of B , two of s , three of H , three of W and two detonation scenarios was simulated once ($2 \times 2 \times 3 \times 3 \times 2 = 72$). The effects of the five inputs can therefore be separated from one another.

A 24-configuration extension populates the interior of the ranges of the three parameters, with six further geometries at two intermediate charge weights in both detonation scenarios ($6 \times 2 \times 2 = 24$). The extension is not balanced, and it is excluded from the analyses that separate the effects of the inputs. The two blocks jointly comprise 96 configurations and span 18 distinct geometries (12 in the core block and 6 in the extension). Two configurations sharing the three parameters of Equation (1) are geometrically similar, and under exact Hopkinson-Cranz scaling their fields coincide when expressed in Z . For each detonation scenario, the four dimensional inputs B , s , H and W therefore reduce to three parameters.

Table 1. Values of the five inputs and the resulting ranges of the three parameters of Equation (1). Core and extension denote the two blocks of simulations.

Parameter	Symbol	Values	Blocks
TNT charge weight (kg)	W	50, 250, 500, 1000, 1500	Core 50, 500, 1500 Extension 250, 1000
Building footprint (m)	B	10, 15, 30	Core 15, 30 Extension 10
Street width (m)	s	5, 8, 12, 20	Core 5, 20 Extension 5, 8, 12
Building height (m)	H	4, 10, 12, 15, 24	Core 4, 12, 24 Extension 10, 15, 24
Detonation type	det	Street (det1), intersection (det2)	Both blocks
Plan-area density (-)	P	0.184-0.735	
Scaled street width ($m/kg^{1/3}$)	$\tilde{s}$	0.44-5.4	
Canyon aspect ratio (-)	H/s	0.20-4.8	

Measured quantities

Every urban run was paired with a matched reference run with identical stages, grids and parameters, but without the buildings, that is a free-field simulation of the same charge. A simulated reference was preferred to the empirical relations so that the comparison reflects the buildings alone. In both runs, the peak overpressure and the positive-phase impulse were recorded in every cell. The urban effect

was expressed as the ratio of the urban to the free-field value at the same point, written P/P_{ff} and i/i_{ff} , the subscript ff denoting the reference run throughout.

In each cell, the urban and the reference values were considered equivalent when their difference fell below a prescribed tolerance. Smaller differences were regarded as insignificant for the engineering purpose of this study. At fixed scaled distance Z , the reference peak overpressure is nearly invariant across the five charge weights, whereas the impulse is invariant only in its scaled form $i/W^{1/3}$. The pressure tolerance was therefore set in absolute terms, so that it selects a single contour in Z , and the impulse tolerance was set on the scaled impulse. Its value is the smallest for which a radius was obtained in every configuration. In peak overpressure a second clause was added, a floor below which the urban overpressure itself is too weak to matter structurally. The floor is anchored in the quantity-distance scale of the International Ammunition Technical Guidelines (IATG) [19], the level below which average damage to unstrengthened buildings remains limited. The two criteria were applied independently, so every quantity below is obtained separately in peak overpressure and in impulse. A cell was classified as equivalent in peak overpressure when Equation (2a) holds, and in impulse when Equation (2b) holds:

$$|P - P_{ff}| < 10 \text{ kPa} \quad \text{or} \quad P < 10 \text{ kPa} \quad (2a)$$

$$\frac{|i - i_{ff}|}{W^{1/3}} < 20 \frac{\text{kPa} \cdot \text{ms}}{\text{kg}^{1/3}} \quad (2b)$$

Near the charge, the buildings alter both the peak overpressure and the impulse. Far from it, the urban values approach those of the reference run. The convergence radius R_{conv} is the distance from the charge beyond which the cells are, globally, equivalent under Equation (2). Beyond R_{conv} the urban load may be taken as the free-field load. Free-field methods, and the safety distances derived from them, may be applied there. Within R_{conv} they may not. The radius was determined separately in peak overpressure and in impulse. It is reported in scaled form, $Z_{conv} = R_{conv}/W^{1/3}$, in $\text{m/kg}^{1/3}$.

The convergence radius was extracted from the computed field in four steps. First, each cell was classified as equivalent or non-equivalent by Equation (2). Cells within the first street, that is close to the charge, were excluded, since the individual buildings dominate there and no radius is meaningful. Second, the field was sampled along 91 directions, one degree apart, from 0 to 90 degrees, each cell assigned to the nearest direction. Third, the cells of each direction were scanned from the outer boundary towards the charge. The scan stopped at the first three consecutive non-equivalent cells, and the radius of the direction was taken there. Three cells were required so that a single anomalous cell could not set the radius. Fourth, the 91 radii r_i , at angular spacing $d\theta = \pi/180 \text{ rad}$, were combined into the radius of a quarter-circle enclosing the same area A :

$$A = \sum_i \frac{1}{2} r_i^2 d\theta; \quad R_{conv} = \sqrt{\frac{4A}{\pi}} \quad (3)$$

In peak overpressure, the sharp tolerance of Equation (2a) made the radius unstable. Where a region of amplification lies close to the tolerance, a small change in geometry moves its cells across the threshold, and the radius jumps abruptly. The difference clause of Equation (2a) was therefore replaced by a smooth transition. Each cell was assigned a probability of being non-equivalent, rising with the pressure difference to one half at the 10 kPa tolerance and one at twice it. The radius of each direction was taken as the expected stopping point of the scan under these probabilities. The floor of Equation (2a) and the impulse criterion of Equation (2b) were kept sharp. The smoothed criterion returns a radius some 17% larger at the median, which is on the safe side for a protective radius. All peak-overpressure radii reported below were obtained with it.

A second quantity states how far a prescribed level of peak overpressure or impulse is carried. For a level occurring at scaled distance Z_{ff} in the free field, R_{urban} is the outermost radius at which the urban field still delivers at least that level. It is extracted along the same directions and combined by Equation (3). Its scaled form Z_{urban} , the amplification factor Λ and the normalised progression ξ are defined by Equations (4) and (5):

$$Z_{urban} = \frac{R_{urban}}{W^{1/3}} \quad (4)$$

$$\Lambda = \frac{Z_{urban}}{Z_{ff}} \quad \xi = \frac{Z_{ff}}{Z_{conv}} \quad (5)$$

$\Lambda > 1$ denotes amplification and $\Lambda < 1$ denotes attenuation of the urban results with respect to the free field case. The normalised progression ξ locates a level within the zone of urban influence, $\xi = 1$ being the convergence radius itself.

Results and Discussion

The Urban Field and the Free Field

The measured quantities defined above were extracted for all 96 configurations. A comparison of the peak overpressure and the impulse shows that the urban layout affects the two differently, and this difference governs the form of every result reported below. The layout amplifies the peak overpressure in some configurations and attenuates it in others, depending on the street width. Over the 96 configurations the two effects nearly cancel. The effect on the impulse does not cancel. It is an amplification, and it is present in almost every configuration. Expressed through the amplification factor of Equation (5), the median is 0.995 in peak overpressure against 1.261 in impulse. On average, the urban layout alters the peak overpressure far less than the impulse.

The peak overpressure registers the single largest arrival. In most of the field this is the leading front. The later reflections are attenuated and usually remain below it, although in a small number of locations a later reflection was found to set the peak. In narrow streets the first reflections from the façades travel with the leading front and raise it, which accounts for the amplification of the peak reported below. The positive-phase impulse is the integration of the pressure, and every reflection contributes to that integral.

Figure 2 presents the ratio fields P/P_{ff} and i/i_{ff} for a narrow street and for a wide one up to 80 m from the charge. Three mechanisms recur. The wave is channelled along the street axis, its lateral expansion inhibited by the parallel façades. Shadowing develops behind the building blocks. Within the layout, the wave reverberates between successive rows, at a more modest excess that is sustained further. All three are pronounced in the impulse field. In the peak overpressure field, they are considerably more local. In both fields the urban influence weakens with distance, until the urban and the free-field values can no longer be distinguished.

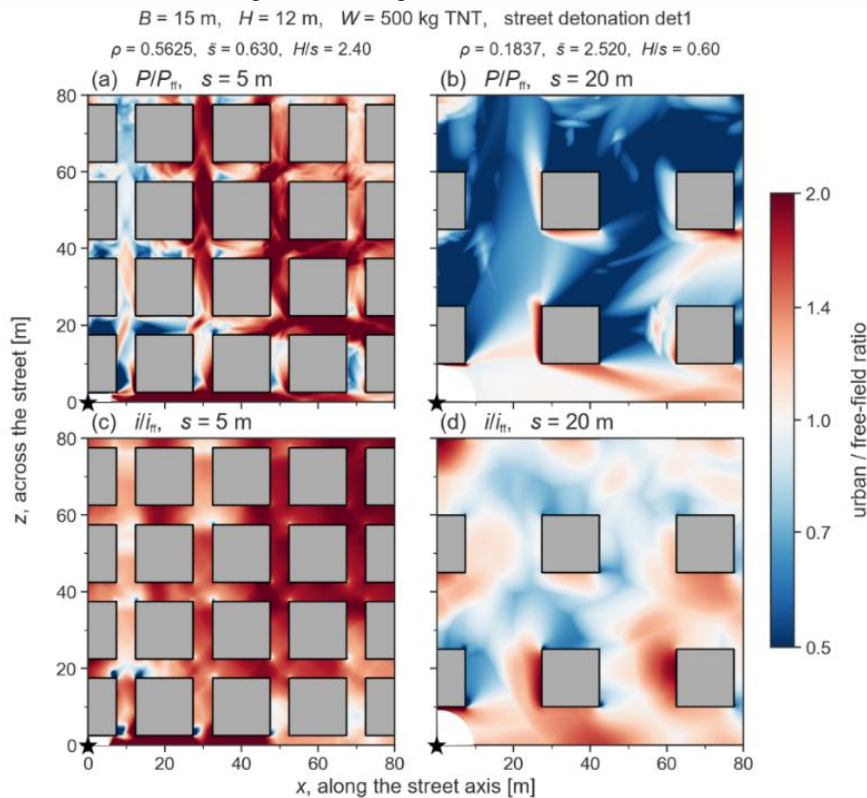


Figure 2. Ratio fields of the urban to the free-field peak overpressure and impulse at ground level, for a narrow and a wide street. Values above unity denote amplification. Cells close to the charge, in the first street, are excluded from the convergence-radius measurement.

The Convergence Radius

The convergence radius defined above was obtained for all 96 configurations, separately in peak overpressure and in impulse, and the two radii differ. The median scaled radius Z_{conv} is $8.99 \text{ m/kg}^{1/3}$ in peak overpressure and $11.61 \text{ m/kg}^{1/3}$ in impulse. At the median, the impulse radius is therefore about 1.3 times the peak overpressure radius ($11.61/8.99$). This ratio depends on the tolerances adopted and is not a physical constant. This is a first engineering consequence of the different effect of the layout on the two quantities: free-field methods become applicable at a larger distance for the impulse than for the peak overpressure. Over all 96 configurations, the scaled radius in peak overpressure does not exceed $13 \text{ m/kg}^{1/3}$. In impulse it exceeds $16 \text{ m/kg}^{1/3}$ in two configurations only, and its largest value is $21.49 \text{ m/kg}^{1/3}$. In peak overpressure this bound follows from the 10 kPa floor of Equation (2a), which the free-field overpressure reaches at a slightly smaller scaled distance. No such floor applies in impulse, and its radius extends farther.

The five inputs were ranked by the size of their effect on the scaled convergence radius, using the core block, in which these effects can be separated. The street width ranks first or second in both peak overpressure and impulse. Its effect is best described through the scaled street width $\tilde{s}$ of Equation (1), that is, the street width relative to the charge, and not the street width in metres. In peak overpressure the charge weight ranks first. In impulse it acts mainly through its interaction with the street width, that is through $\tilde{s}$.

The scaled street width also marks a threshold. Where the street is narrow relative to the charge, the canyon confines the wave and channels it along the street. Where it is wide, the blocks obstruct the wave instead, and where it is very wide the building height only weakly affects the peak-overpressure radius. The transition lies near $\tilde{s} = 1 \text{ m/kg}^{1/3}$, and it was measured rather than assumed. This value was proposed in advance, and the results are consistent with it. Below it roughly three configurations in four amplify the peak overpressure. Above it amplification ceases to be systematic, and where it persists it is weak. In peak overpressure, the effect of the canyon aspect ratio on the radius, measured by the slope $dZ_{\text{conv}}/d(H/s)$, also reverses at this value: below it, taller buildings increase the radius, and above it they reduce it in nearly every case. In impulse, no such reversal occurs at any street width. Figure 3 presents the effect of building height on the radius and its reversal at $\tilde{s} = 1 \text{ m/kg}^{1/3}$.

The two quantities differ in a further respect. Among the three parameters of Equation (1), the impulse radius depends mainly on the plan-area density ρ , and the dependence is a smooth power law. A denser layout returns more reflection and gives a larger radius. In peak overpressure, the interactions between the inputs, that is their combined effects, are as large as the effects of each input alone, and no smooth power law describes the radius.

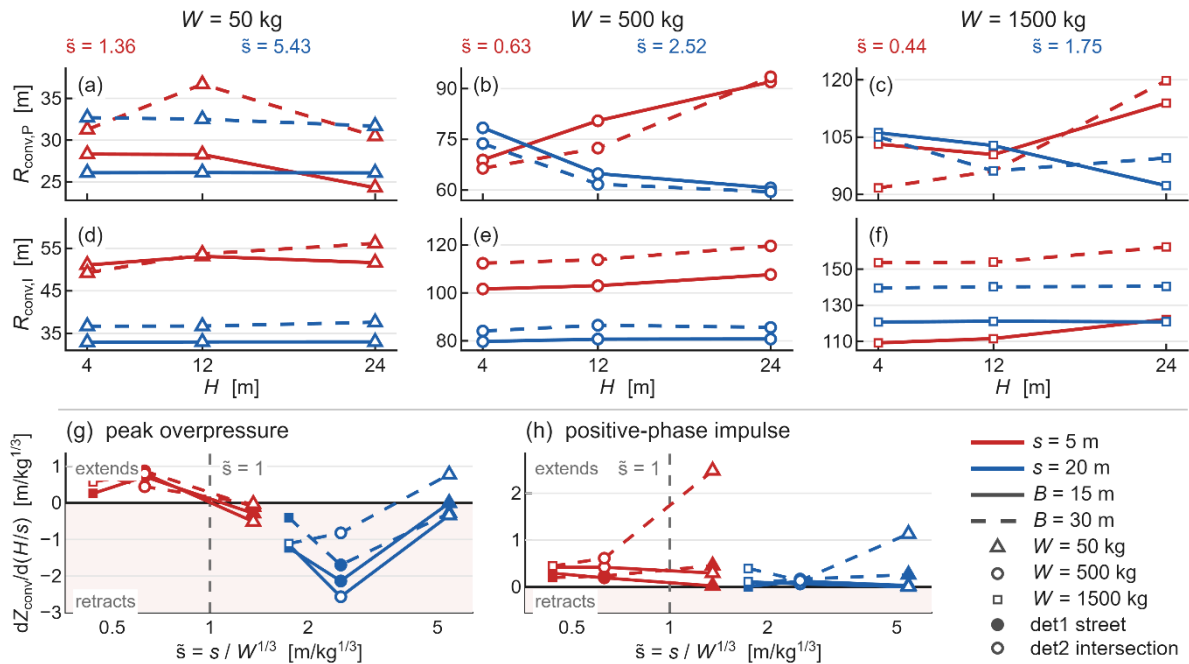


Figure 3. Effect of building height on the convergence radius, and its reversal at $\tilde{s} = 1 \text{ m/kg}^{1/3}$. (a) to (f) Convergence radius against building height at three charge weights, only

the height varying along each line. (g, h) Height sensitivity $dZ_{conv}/d(H/s)$ for the 24 series of the core block in which only the height varies.

Closed-form prediction of the convergence radius

The difference between the peak overpressure and the impulse, described above, dictates the form of the prediction. Closed-form expressions for the scaled convergence radius were fitted to the measured values, separately for peak overpressure and impulse and for each detonation scenario. In peak overpressure the radius is written as a sum of terms, two of which reverse at a stated geometry:

$$Z_{conv,P} = C_0 + C_1 \cdot \tilde{s} + C_2 \cdot \rho \cdot (\tilde{s} - a) + C_3 \cdot \sqrt{\rho} \cdot \left(\frac{H}{s}\right) \cdot \left(\frac{W^3}{s} - 1\right) \quad (6)$$

The density term, with coefficient C_2 , reverses at $\tilde{s} = a$, a fixed constant of each detonation scenario. The canyon term, with coefficient C_3 , reverses at $\tilde{s} = 1 \text{ m/kg}^{1/3}$, the threshold identified above. In impulse neither reversal occurs. The impulse radius is accordingly written as a power law in the three parameters. A single logarithmic correction in $\tilde{s}$ is added, since the radius peaks at an intermediate scaled street width:

$$Z_{conv,I} = k \cdot \rho^p \cdot \left(\frac{H}{s}\right)^q \cdot \tilde{s}^r \cdot e^{r_2 \cdot (\ln \tilde{s})^2} \quad (7)$$

The difference between the two forms is physical and not a fitting preference. Fitting either form to the other quantity increases the prediction error by three to four percentage points, in both detonation scenarios. The fitted coefficients are given in Table 2. Both equations are numerical-value expressions, with $\tilde{s}$ entered in $\text{m/kg}^{1/3}$ and Z_{conv} returned in $\text{m/kg}^{1/3}$.

The formulas were validated by leave-one-geometry-out (LOGO) cross-validation. All configurations of one geometry were withheld in turn and the coefficients refitted on the remainder, so the error reported is that of a geometry the fit has never seen. Figure 4 presents the outcome. The mean absolute percentage error (MAPE) of the predicted radius remains below 10% in both quantities. The largest errors occur at the corners of the sampled ranges, where a withheld geometry has no neighbouring geometry from which to interpolate. A supporting test, with 500 random splits of the configurations, returned median errors of 8.8% in peak overpressure and 7.2% in impulse. In most of these splits, a withheld configuration has another configuration of the same geometry in the fitting set, so this test checks mainly the effect of the charge weight.

The formulas interpolate within the ranges of Table 1 and are not to be applied outside them. A narrower range can be delimited, with ρ between 0.31 and 0.56, $\tilde{s}$ between 0.50 and $5.4 \text{ m/kg}^{1/3}$ and H/s not exceeding 3. Within it every withheld prediction falls below 20% error, in both quantities. One further condition applies. The periodic description becomes representative only over several blocks. A prediction should accordingly be sought only at a radius exceeding roughly 1.5 unit cells, $R \geq 1.5(B+s)$. This condition can be checked in advance.

Table 2. Fitted coefficients of the closed-form convergence radius. The upper block gives Equation (6) in peak overpressure and the lower block Equation (7) in positive-phase impulse, each row from one fit to all configurations of its detonation scenario. The coefficients are specific to the tolerances of Equation (2). The value a , at which the C_2 term reverses, is fixed and not fitted.

Equation (6), Peak Overpressure	C0	C1	C2	C3	a
Street detonation, det1	9.063	-0.645	2.018	0.591	1
Intersection detonation, det2	11.858	-1.375	2.782	0.776	2
Equation (7), Positive-Phase Impulse	k	p	q	r	r ₂
Street detonation, det1	14.936	0.203	0.024	0.065	-0.101
Intersection detonation, det2	14.602	0.176	0.079	0.125	-0.089

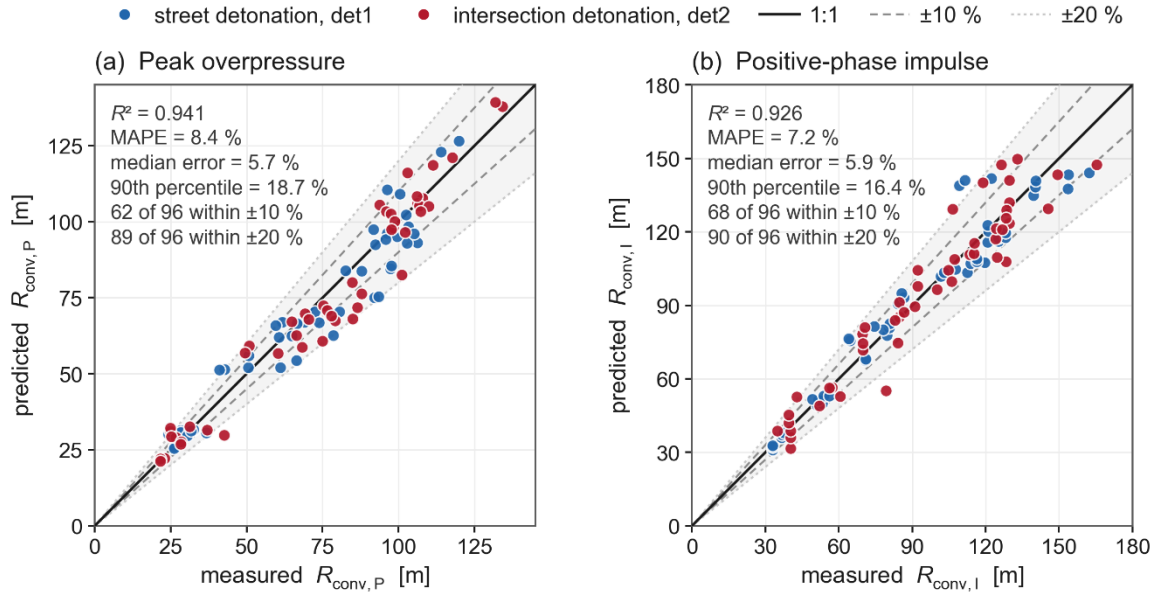


Figure 4. Predicted against measured convergence radius under LOGO cross-validation, one point per configuration. Only predictions for withheld geometries are shown.

With Equations (6) and (7), the convergence radius, which bounds the urban influence, is predictable in closed form. The peak overpressure and impulse inside the radius are not yet predictable in this way. Closed-form equations have been fitted there and are undergoing LOGO validation. In terms of Λ and ξ of Equation (5), the amplification in peak overpressure is largest near the charge and decays outwards. This is consistent with interference between the direct and the reflected fronts. In impulse no such dependence was found. Each configuration sets a single amplification factor that holds at every distance within the radius.

Summary

A parametric study of blast loading in an idealised urban environment has been reported. A total of 96 three-dimensional CFD simulations were conducted in ViperBlast over a periodic array of identical rectangular buildings, each paired with a matched free-field reference run. The array was described by three parameters, the plan-area density ρ , the scaled street width $\tilde{s}$ and the canyon aspect ratio H/s . The charge weight enters the fitted relations through the Hopkinson length, so they may be applied to charge weights between those simulated.

The principal finding is that the urban layout affects the peak overpressure and the impulse differently. The layout alters the peak overpressure only modestly on average, and it amplifies the positive-phase impulse in almost every configuration. The impulse is therefore the quantity most affected by the urban environment.

The urban influence falls below the prescribed tolerances at a finite distance, the convergence radius. This radius is predictable in closed form from the three parameters, but two structurally different expressions are required. An additive form describes the peak overpressure and captures the reversal from channelling to blocking at $\tilde{s} = 1 \text{ m/kg}^{1/3}$. A power law describes the impulse, in which this reversal does not occur. Under LOGO validation the mean absolute percentage error of the predicted radius remains below 10% in both quantities.

The results are provisional and subject to several limitations. The predicted radius is a single value, whereas the urban influence varies with direction. The idealised layout is uniform and effectively unbounded, whereas real urban layouts are not. The coefficients are those of the tolerance of Equation (2) alone, and a different tolerance would change them. Work now proceeds on the loads within the radius, on the dependence of the amplification factor on direction, and on filling the corners of the parameter space that the present design sampled sparsely. Experimental validation remains the necessary complement to all three.

Acknowledgement

This work was supported by the PMRI – Peter Munk Research Institute – Technion.

References

- [1] Norville, H.S., Harvill, N., Conrath, E.J., Shariat, S., and Mallonee, S., *Journal of Performance of Constructed Facilities* 13:50–56 (1999).
- [2] Hopkinson, B., “British Ordnance Board Minutes,” Ordnance Board Report No. 13565, 1915.
- [3] Cranz, C., *Lehrbuch der Ballistik*, Springer, 1926.
- [4] Kingery, C.N., and Bulmash, G., “Airblast parameters from TNT spherical air burst and hemispherical surface burst,” US Army Armament Research and Development Center, Ballistic Research Laboratory Report No. ARBRL-TR-02555, 1984.
- [5] Smith, P.D., and Rose, T.A., *Progress in Structural Engineering and Materials* 8:16–28 (2006).
- [6] Denny, J., Langdon, G., Rigby, S., Dickinson, A., and Batchelor, J., *International Journal of Protective Structures* 15:3–22 (2024).
- [7] Gajewski, T., and Sielicki, P.W., *Shock Waves* 30:385–394 (2020).
- [8] Ratcliff, A., Rigby, S., Clarke, S., and Fay, S., *Applied Sciences* 13:5349 (2023).
- [9] Rose, T.A., and Smith, P.D., *International Journal of Impact Engineering* 27:359–376 (2002).
- [10] Fouchier, C., Laboureur, D., Youinou, L., Lapébie, E., and Buchlin, J.M., *Journal of Loss Prevention in the Process Industries* 49:248–265 (2017).
- [11] Ripley, R.C., Cloney, C., Donahue, L., and Zhang, F., *Proceedings of the 22nd International Symposium on Military Aspects of Blast and Shock*, Bourges, France, November 2012.
- [12] Hedhli, M., Lapébie, E., Marinasy, E., and Sochet, I., *Structures Under Shock and Impact XVII*, Edinburgh, UK, June 2025, pp. 51–62.
- [13] Dennis, A.A., and Rigby, S.E., *International Journal of Protective Structures* 15:455–483 (2024).
- [14] Chester, A., Hazael, R., and Critchley, R., *International Journal of Protective Structures* 16:825–852 (2025).
- [15] Viper Applied Science, “VIPER::Blast,” 2024.
- [16] Smith, P.D., Rose, T.A., Krahe, S.L., and Franks, M.A., *Proceedings of the Institution of Civil Engineers - Structures and Buildings* 156:359–365 (2003).
- [17] Dib, N., Zéhil, G.P., and Rigby, S., *Journal of Fluids and Structures* 115:103787 (2022).
- [18] Angelides, S.C., Morison, C., Burgan, B.A., Kyprianou, C., Rigby, S.E., and Tyas, A., *Structures* 58:105619 (2023).
- [19] United Nations Office for Disarmament Affairs, “IATG 02.20: Quantity and Separation Distances, 3rd Edition,” 2021.